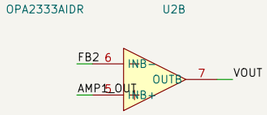
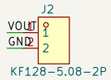


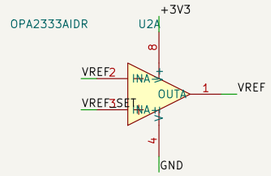
NO OUTPUT ISOLATION RESISTOR: OPA2333 Fig.15's mid-30 % overshoot into 1 nF is a UNITY-GAIN figure.

This stage runs at noise gain 3.49 - simulated 6.6 % into the 1 nF of Q8, ~63 deg phase margin.





SIGNAL CHAIN J1 -> U1 (G1 = 39.90) -> U2B (G2 = 3.49) -> J2



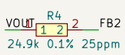


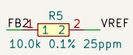


PEDESTAL R2/R3 off +3V3 = 0.2519 V, C4 on the DIVIDER node, U2A buffers it for U1 REF

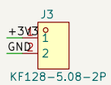
AD8226's 2 ohm REF rule: met at DC and low frequency (B9: 0.43 mohm DC, 0.17 ohm at 60 Hz),

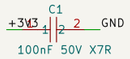
crossed above ~700 Hz - harmless, Q1's common mode is STATIC. Numbers in the box.

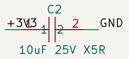


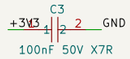


STAGE-2 GAIN SET R5 RETURNS TO /VREF, NOT GND - that is what makes the pedestal EXACT









POWER +3V3 arrives at J3; C2 10uF bulk at J3; C1/C3 100nF at pin 8 of U1/U2 (AD8226 Fig.61, OPA2333 p15)





bb-amp - BRIDGE FRONT END ON ONE 3.3 V RAIL (architecture/blocks.md)

TRANSFER FUNCTION
 Vout = Vref + G1*G2*Vdiff G1*G2 = 139.2 V/V, Vref = 0.252 V
 Vout(-1 mV) = 0.113 V Vout(0) = 0.252 V Vout(+20 mV) = 3.037 V
 +25 mV overload clips at (V+) - 0.05 V and recovers, no fold-back.

GAIN SPLIT G1 * G2 = 39.90 * 3.49
 G1 = 1 + 49.4k/RG = 1 + 49.4k/1.27k = 39.90 U1 AD8226, R1 across pins 2-3
 G2 = 1 + R4/R5 = 1 + 24.9k/10.0k = 3.49 U2B, OPA2333 half B
WHY IT IS SPLIT: the AD8226's INPUT VOLTAGE RANGE section (Eq.1-3, Table 8)
limits the internal 1st/2nd-stage nodes, not the pins. Eq.2 ceiling on
|G*Vdiff| and the gain it allows at 20 mV FS (blocks.md Ruling 2) - note
which Vcm each row belongs to:
 Vcm 1.65 V (Q1), rail -5 %, 0 degC 1.49 V G <= 75
 Vcm 1.7325 V, rail -5 % AND excitation +5 % 1.33 V G <= 66
1.65 V is the stated operating point; the 1.33 V / G <= 66 pair is the
WORST CORNER, not the nominal common mode. Either way a single stage at
G = 139 saturates INSIDE the part while its input and its output both
still look legal. The data sheet's own remedy: less gain in the in-amp,
the rest later in the chain.

PEDESTAL Vref = 3.3 V * R3/(R2+R3) = 3.3 * 10.0k/131.0k = 0.2519 V
 C4 filters the DIVIDER node (Thevenin 9.24k -> 172 Hz corner), never the
 buffer output - a follower into a capacitive load peaks.
 U2A buffers it. AD8226 Fig.59 draws a bare divider into REF, crosses it
 out and marks it INCORRECT: source impedance at REF must stay below
 2 ohm, or the +IN path is amplified by 2(50k + Rref)/(100k + Rref) and
 CMRR collapses (9.24k there = ~8.5 % error = ~21 dB of CMRR left).

HOW WELL THE BUFFER MEETS THAT RULE IS MEASURED, NOT FLAT (bench B9):
 DC 0.43 mohm 60 Hz 0.17 ohm 1 kHz 2.86 ohm 41 kHz 116 ohm
Met through DC and the low frequencies where the AD8226 specifies CMRR;
EXCEEDED above about 700 Hz (350 Hz on the pessimistic ro = 2 kohm
bracket). Not a defect here - Q1 fixes the common mode at a STATIC
1.65 V with nothing at 1 kHz to convert, and the residual common-mode
gain error is 90.8 dB, at or above the part's own 90 dB min at 5 kHz.
A board with a MOVING common mode would have to rederive this.

*** R5 RETURNS TO /VREF, NOT TO GROUND ***
 U2B's + input sits at V1 = Vref + G1*Vdiff.
 With R5 to /VREF: Vout = V1 + (V1 - Vref)*R4/R5 = Vref + G1*G2*Vdiff.
 The output pedestal is therefore EXACTLY Vref - no matched network, no
 second reference node - and a shift of Vref lands on U1 REF and on the
 stage-2 return with OPPOSITE sign, so the reference's own drift and noise
 cancel to first order and survive only as the gain-1 pedestal term.
 With R5 to GND the pedestal would become G2*Vref and every bit of
 reference error would be amplified by G2. A different, worse board.

/VREF IS ONE NODE WITH THREE LOADS: U2A OUT (1), U1 REF (6), R5.

SINGLE SUPPLY: U1 pin 5 (-VS) and U2 pin 4 (V-) go to GND. The RRIO OPA2333
(30 mV typ from either rail) is the part that faces the rails, not the AD8226
(0.10 V guaranteed floor) - the same spec that makes a grounded REF unusable.

ACCURACY, RECORDED NOT SMOOTHED (blocks.md Ruling 3): Q7's 5 uV RTI over
0-50 degC is 10 ppm of full scale per degC and is NOT met - 13.9 uV typ /
56.4 uV max offset drift, plus 30-87 uV of gain drift at full scale. No
3.3 V-capable part reaches it; zero and span are calibrated downstream.
THE TERM blocks.md's TABLE OMITS - RAIL SENSITIVITY. The pedestal is a
RATIO of the rail (0.2519/3.3 = 7.63 %) and reaches the output at gain 1,
so dVout/dVs = 0.0763 V/V (bench B6 measures exactly that) = 548 uV RTI
per volt of rail. So 0.28 % of rail movement AFTER calibration - about
9.1 mV on 3.3 V - eats the whole 5 uV budget on its own, which makes it
the DOMINANT post-calibration term, ahead of the 1.1 uV R2/R3 ratio-TCR
the table does count. Not an error and not a scope miss (a static
pedestal calibrates out; a voltage reference is excluded at block-only) -
it is a requirement on the +3V3 SOURCE, stated here so it is not lost.

NO OUTPUT ISOLATION RESISTOR, AND THAT IS A RESULT, NOT AN OMISSION. The
OPA2333's -mid-30 % overshoot into 1 nF (Fig.15) is a UNITY-GAIN figure; this
stage runs at noise gain 3.49. Simulated into the 1 nF that Q8 allows, the
as-built stage overshoots 6.6 % with about 63 deg of phase margin, so nothing
needs isolating - and at block-only scope, conditioning the data sheet does
not require is out of scope.

ai-ee

Sheet: /

File: bb-amp.kicad_sch

Title: bb-amp bridge front end AD8226 x OPA2333, G=139.2, 3.3V

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